

Software Requirement Specification (SRS)

Project Name: DigiExam **Version:** 1.0 **Date:** January 19, 2026

1. Introduction

1.1 Purpose

The purpose of this project is to develop a secure, web-based examination platform tailored for Police Academy cadets. The system aims to digitize the evaluation process using decentralized storage (IPFS), AI-driven performance analysis, and strict "Classroom Mode" proctoring to maintain exam integrity.

1.2 Scope

The application will facilitate:

- Secure administration of timed exams with randomized questions.
 - Decentralized storage of exam assets (images/docs) via IPFS.
 - Automated proctoring (Anti-cheat mechanisms).
 - AI-powered feedback for student learning.
 - Automated reporting to supervising officers via email.
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2. Technology Stack

- **Frontend Framework:** Next.js (React)
 - **UI/Styling:** Tailwind CSS
 - **Database:** MongoDB (User profiles, Logs, Exam Metadata)
 - **File Storage:** IPFS (InterPlanetary File System) for immutable storage of large exam files/question papers.
 - **AI Integration:** LLM API (e.g., OpenAI/Gemini) for remedial feedback.
 - **Communication:** Nodemailer/SMTP for automated reporting.
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3. User Roles

3.1 Administrator (Instructor/IT Cell)

- Full access to system configurations.
- Manages Question Banks and Supervisor lists.
- Controls the "Session Gate" (OTP generation) to start exams.

3.2 User (Cadet)

- Restricted access.
 - Can only attempt exams when authorized via Session OTP.
 - View post-exam analysis and AI feedback.
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4. Functional Requirements

4.1 Authentication Module

- **Credential Login:** Users and Admins log in using a standard Employee ID/Email and Password.
- **Session Security:** JWT (JSON Web Tokens) used to manage session states.

4.2 Admin Module & Exam Management

- **Question Bank Management:**
 - Admin uploads exam documents/images.
 - **System Action:** Files are uploaded to **IPFS**; only the Content Identifier (CID) hash is stored in MongoDB.
- **Supervisor Allocation:** Admin assigns a specific "Reporting Supervisor" (Respected Person) email ID to each exam.
- **Session OTP (The Gatekeeper):**
 - To start an exam batch, Admin clicks "Generate Access Code."
 - System generates a 6-digit OTP.
 - **Constraint:** OTP validity expires strictly after **5 minutes**. Admin can regenerate if necessary.

4.3 Exam Interface (Cadet Module)

- **Exam Entry:**
 - Cadet selects an exam.
 - System prompts for the "Session Access Code" (generated by Admin).
 - If the code is valid and active, the exam loads.
- **Randomization Engine:**
 - Questions are fetched and shuffled randomly for *each* user to prevent copying.
- **User Interface:**
 - Sidebar navigation for questions.
 - Countdown timer.

4.4 Proctoring & Anti-Cheat System (Critical)

- **Full-Screen Enforcement:** The exam interface forces the browser into Full-Screen mode. Questions are not visible otherwise.
- **Violation Monitoring (The 5-Strike Rule):**
 - The system detects: Tab switching, minimizing browser, or exiting full screen.
 - **Action:**
 - Strikes 1-4: User receives a warning toast notification.
 - Strike 5: System **immediately terminates** the session and auto-submits the exam.
 - Incident is logged in the database.

4.5 Post-Exam & AI Analysis

- **Grading:** Immediate score calculation upon submission.
- **AI Remediation:**
 - System identifies incorrect answers.
 - Queries AI Engine: *"Explain why option [User Answer] is wrong and [Correct Option] is right based on this question."*
 - Displays the AI explanation to the Cadet for learning purposes.

4.6 Reporting Module

- **Automated Dispatch:**
 - Upon exam completion, the system generates a result summary.
 - **Action:** Automatically sends an email to the assigned Supervisor's email ID containing:
 - Cadet Name/ID.
 - Score obtained.
 - Time taken.
 - Violation Count (if any).
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5. System Workflow Summary

1. **Preparation:** Admin creates exam, uploads assets to IPFS, and sets the Supervisor email.
2. **Assembly:** Cadets log in with credentials and wait in the lobby.
3. **Authorization:** Admin projects/shares the **Session OTP** (Valid for 5 mins).
4. **Exam Start:** Cadets enter OTP. Browser enters Full Screen.
5. **Execution:** Cadets answer questions. System monitors for tab-switching (5 strikes = Auto-fail).
6. **Submission:** Timer ends OR User submits OR Violation limit reached.
7. **Feedback:** Cadet sees score + AI explanations for errors.
8. **Reporting:** Supervisor receives the result via email immediately.

6. Non-Functional Requirements

- **Performance:** The randomization algorithm must run client-side or efficiently server-side to ensure no lag during exam load.
- **Reliability:** If the user disconnects, the timer continues running server-side.
- **Data Integrity:** Exam papers stored on IPFS cannot be tampered with (immutable).